

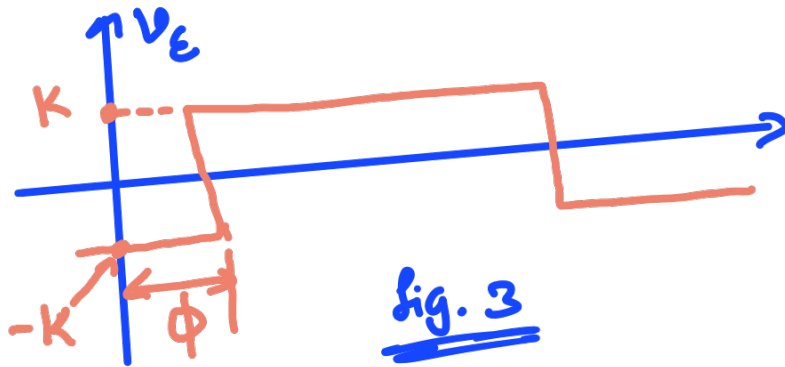
①

In this question, we are not concerned with switching frequency component & its harmonics. we are only concerned with fundamental frequency component & harmonics of fundamental frequency which are wayy lower than switching frequency.

$$\therefore (V_{Ao})_{ideal} = (\hat{V}_{Ao}),$$

as explained in the question and shown in Fig. 2B

V_E can be plotted as,



it is said that value of K is such that, volt-sec area per half cycle is equal to pulses shown in Fig. 2B when, T_s = switching period & t_d = dead time.

$$\therefore K = V_d \times \frac{t_d}{T_s}$$

the fourier series of the waveform shown in above fig. 3 can be easily obtained,

$$V_E(t) = \sum_{h=1,3,5,\dots} \frac{hK}{\pi h} \sin[h(\omega_m t - \phi)]$$

where, $\omega_m = 2\pi f_m$, $f_m =$ modulating wave frequency
or
fundamental frequency
and $h = 1, 3, 5, \dots$ [odd harmonics only].

$$\therefore V_E(t) = \sum_{h=1,3,5,\dots} \frac{h}{\pi h} \cdot V_d \cdot \frac{t_\Delta}{T_s} \cdot \sin[h(\omega_m t - \phi)]$$

$$\therefore V_E(t) = \sum_{h=1,3,5,\dots} \frac{h}{\pi h} V_d t_\Delta f_s \sin[h(\omega_m t - \phi)]$$

$$\therefore (V_{AO})_{\text{actual}} = (V_{AO})_{\text{ideal}} - V_E$$

$$\therefore (V_{AO})_{\text{actual}} = \frac{m_a V_d}{2} \sin(\omega_m t) - \sum_{h=1,3,5,\dots} \frac{h V_d t_\Delta f_s}{\pi h} \sin[h(\omega_m t - \phi)]$$

$$(V_{AO})_{\text{actual}} = \left[\frac{m_a V_d}{2} \sin(\omega_m t) \right] - \left[\frac{h V_d t_\Delta f_s}{\pi} \sin(\omega_m t - \phi) \right]$$

$$- \sum_{h=3,5,\dots} \frac{h V_d t_\Delta f_s}{\pi h} \sin[h(\omega_m t - \phi)]$$

lower order harmonic of
fundamental frequency

② a, d

③

a

given that, rising edge delay = $h \cdot T_s$

$$\therefore \text{dead time} = 0.04 T_s$$

$$\therefore \text{maximum duty to any switch, } (d_k)_{\max} = \frac{T_s - 0.04 T_s}{T_s}$$

$$(d_k)_{\max} = 0.96$$

$\therefore$ for $(d_k)_{\max} = 0.96$, $(m)_{\max}$ for linear modulation range can be found out from (1)

$$(m)_{\max} = (0.96 \times 2) - 1 = \underline{\underline{0.92}}$$

b

In assignment 1, we already looked at maxima & minima points for (2), so one of the maxima point occur at $\omega t = 60^\circ$

$$\therefore d_1 = \frac{1}{2} + \frac{m}{2} \sin \omega t + \frac{m}{2} \sin(3\omega t)$$

then due to dead time, $(d_1)_{\max} = 0.96$.
at $\omega t = 60^\circ$

$$\therefore 0.96 = \frac{1}{2} + \frac{m}{2} \sin 60 + \frac{m}{12} \sin(3 \times 60)$$

$$(m)_{\max} = 1.0623$$

(4)

(i)

$$\bar{I} = \frac{10 \times 10^3}{\sqrt{3} \times 415 \times 0.9} \angle -\cos^{-1}(0.9)$$

$$\bar{I} = 15.46 \angle -25.842^\circ \text{ A}$$

$$I_{\text{rms}} = 15.46 \text{ A}$$

(ii) $\bar{V}_{\text{inductor}} = j \times 2 \times \pi \times 50 \times 1 \times 10^{-3} \times 15.46 \angle -25.842^\circ$

$$\bar{V}_{\text{inductor}} = 4.857 \angle 64.158^\circ$$

$$(V_{\text{inductor}})_{\text{rms}} = 4.857 \text{ V.}$$

$$(iii) \quad \bar{V}_{inv} = \bar{V}_{inductor} + \bar{V}_{grid}$$

$$\bar{V}_{inv} = (4.857 \angle 64.158^\circ) + \left(\frac{415}{\sqrt{3}}\right) \angle 0^\circ$$

$$\bar{V}_{inv} = 241.76 \angle 1.036^\circ$$

$$\boxed{(V_{inv})_{rms} = 241.76 V}$$

(iv)

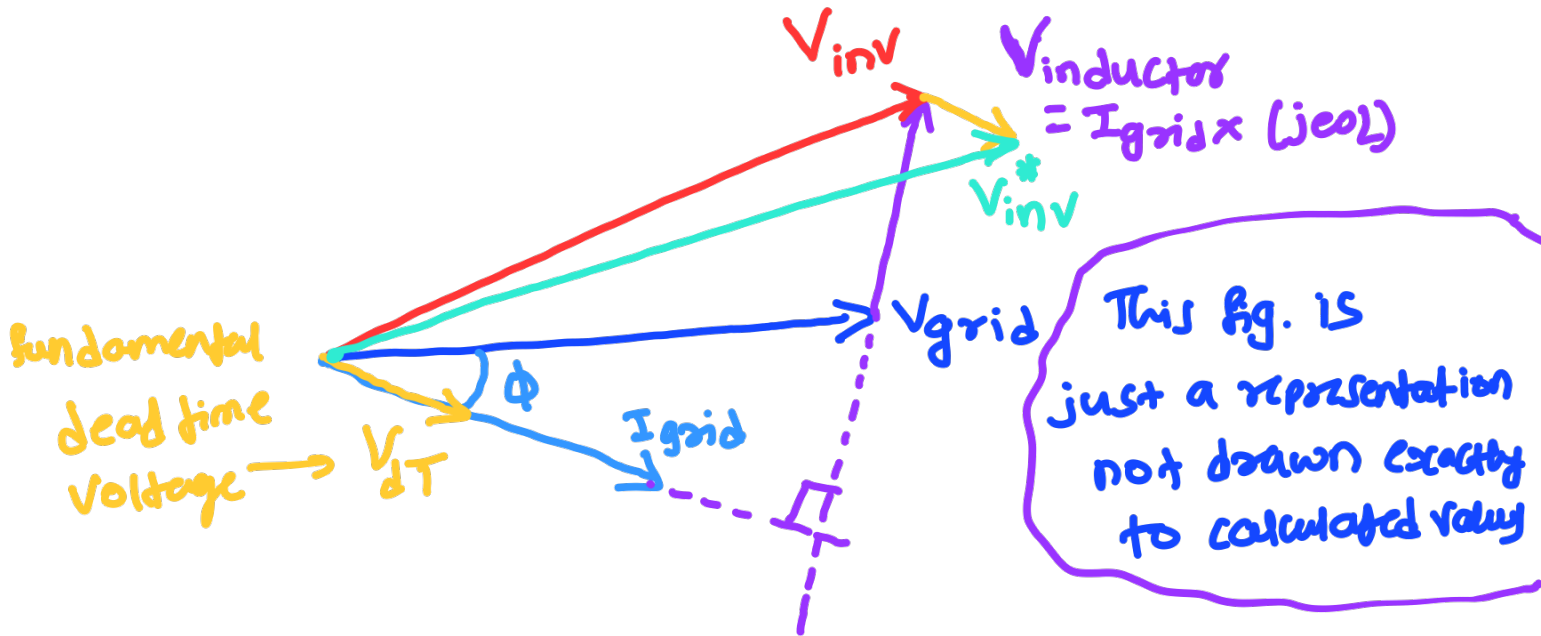
as can be seen from question ① solution, when fundamental current is lagging the ideal $(V_{Ao})_{ideal}$ and ripple is small, the fundamental of error voltage is in phase with fundamental current.

$$\therefore (V_E)_{fundamental} = \frac{4V_d t_d}{\pi T_s} \sin(\omega t - \phi)$$

$$(\bar{V}_E)_{fund} = \frac{4V_d t_d}{\sqrt{2} \pi T_s} \angle -25.842^\circ$$

$$(\bar{V}_E)_{fund} = \frac{4 \times 1000 \times 0.04}{\sqrt{2} \pi} \angle -25.842^\circ$$

$$(\bar{V}_E)_{fund} = 36.0127 \angle -25.842^\circ$$



$\bar{V}_{inv}$ that was calculated in (iii) is after the dead time has occurred. Thus in order to get the reference we have to add $(\bar{V}_E)_{fund}$ to $(\bar{V}_{inv})$

$$\therefore \bar{V}^* = \bar{V}_{inv} + (\bar{V}_E)_{fund}$$

$$\bar{V}^* = 241.76 \angle 1.036^\circ + 36.0127 \angle -25.842^\circ$$

$$\bar{V}^* = 274.366 \angle -2.3659^\circ$$

$\therefore$ modulating signal

$$m_{r,a} = \frac{\sqrt{2} \times 274.366}{\left(\frac{1000}{2}\right)} \sin(\omega_0 t - 2.3659)$$

$$\therefore m_{r,a} = 0.7760 \sin(\omega_0 t - 2.3659)$$

$$m_{r,b} = 0.7760 \sin(\omega_0 t - 122.3659)$$

$$m_{r,c} = 0.7760 \sin(\omega_0 t - 242.3659)$$

Similar question was asked in Quiz-2
without dead time. compare the modulating
waveform with and without dead time.

5

i) in phase voltage, there will be

$\left[2 \times [\text{no. of cells}] + 1 \right]$ levels will be present

$\therefore (2 \times 5) + 1 = 11$ levels in phase voltage.

ii)

Peak fundamental line voltage = $[\text{no. of cell}] \times \text{max } V_{dc}$

$$= \sqrt{3} \times 5 \times 0.9 \times 2000 = \underline{15,588.46 \text{ V}}$$

iii) device voltage rating = 2000 volts.

iv) load current = $\frac{\text{apparent power}}{\sqrt{3} \times (\text{line voltage rms})}$

$$\begin{aligned} \text{apparent power} &= \sqrt{P^2 + Q^2} = \sqrt{5^2 + 5^2} \text{ MVA} \\ &= 7.07 \text{ MVA} \end{aligned}$$

$$\text{line voltage}_{\text{rms}} = \frac{\sqrt{3} \times [\text{no. of cells}] \times \text{max } V_{dc}}{\sqrt{2}}$$

$$= \frac{\sqrt{3} \times 5 \times 0.9 \times 2000}{\sqrt{2}}$$

$$= 11022.704 \text{ V.}$$

$$\therefore \text{load current}_{\text{rms}} = \frac{7.07 \times 10^6}{\sqrt{3} \times 11022.704} = 370.314 \text{ A}$$

$$\begin{aligned} \text{v) device current rating} &= 370.314 \times \sqrt{2} \text{ A (Peak)} \\ &= \underline{\underline{523.7036 \text{ A (Peak)}}} \end{aligned}$$

⑥ a) unequal loss distribution.

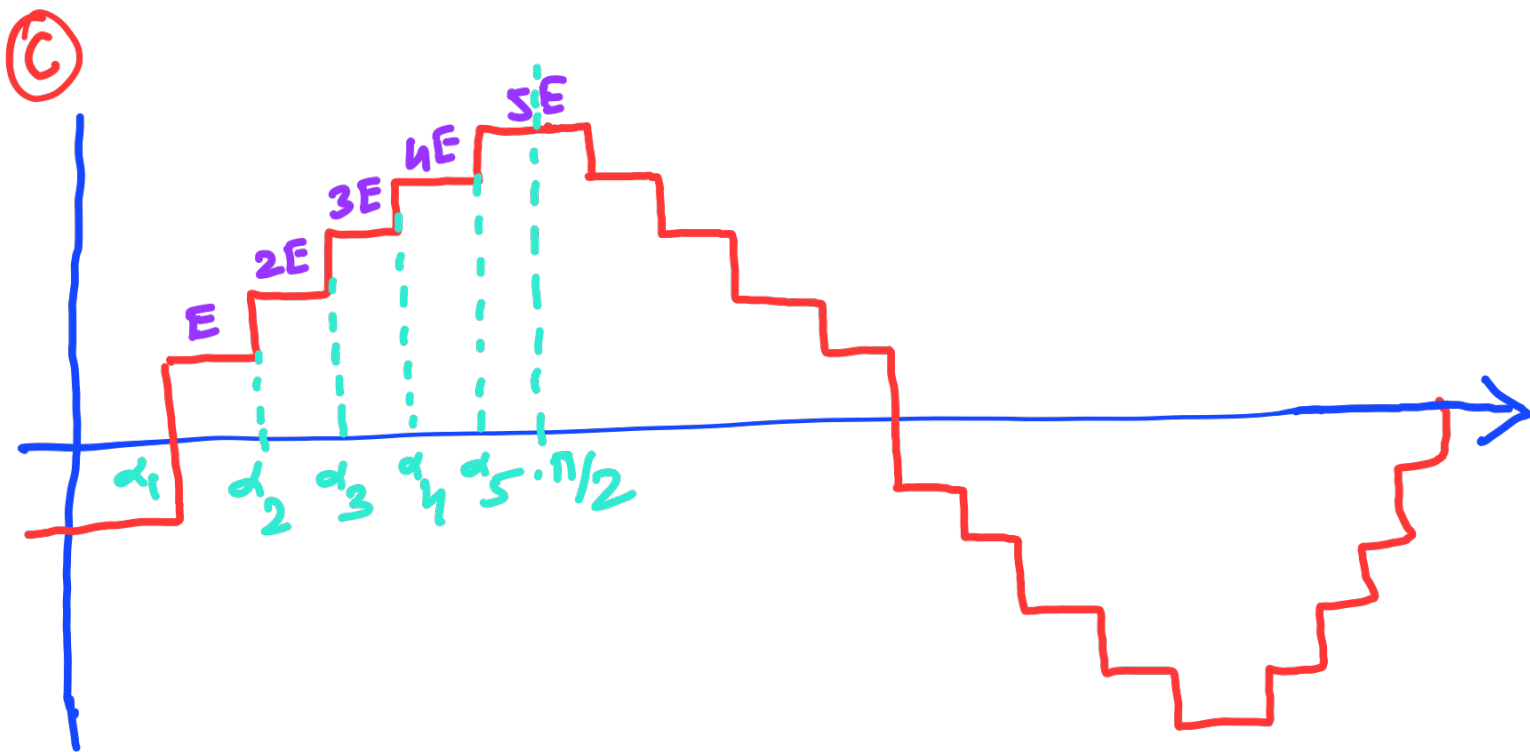
⑦ c) $20 V_{dc}$

$$\text{⑧ (a) } V_{ph, \text{Peak}} = \sqrt{2} \times \frac{3000}{\sqrt{3}} = 2449.49 \text{ V}$$

$$[\text{no. of cells}] \times \text{Max } V_{dc, \text{min}} = 2449.49 \text{ V}$$

$$V_{dc, \text{min}} = \frac{2449.49}{1 \times 5} = 489.89 \text{ V}$$

$$\text{⑧ (b) } [2 \times (\text{no. of cells}) + 1] = (2 \times 5) + 1 = 11 \text{ levels.}$$



it is said that $\alpha_1, \alpha_2, \alpha_3$ & α_4 are uniformly distributed among themselves and also with extremities. It is also said that both extremities are not the candidates for switching instants.

Then,

$$\alpha_1 = 15^\circ, \alpha_2 = 30^\circ, \alpha_3 = 45^\circ, \alpha_4 = 60^\circ, \alpha_5 = 75^\circ$$

It is already derived in class that,

$$b_n = \frac{8E}{2n\pi} \cos(n\alpha), \text{ where } n \rightarrow \text{odd}$$

i) fundamental amplitude will be,

$$= \frac{8E}{2\pi} [\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3 + \cos \alpha_4 + \cos \alpha_5]$$

$$= \frac{8 \times 500}{2\pi} [\cos(15^\circ) + \cos(30^\circ) + \cos(45^\circ) + \cos(60^\circ) + \cos(75^\circ)]$$

$$= 2099.4937 \text{ V. (Peak).}$$

$$\Rightarrow \frac{2099.4937}{\sqrt{2}} \Rightarrow \underline{\text{rms value}}$$

ii) 3rd harmonic rms

$$= \frac{8E}{2 \times 3 \times \pi \times \sqrt{2}} [\cos 3\alpha_1 + \cos 3\alpha_2 + \cos 3\alpha_3 + \cos 3\alpha_4 + \cos 3\alpha_5]$$

$$= -256.156 \text{ V}$$

11y

$$(V_5)_{\text{rms}} = 13.65 \text{ V}$$

$$(V_7)_{\text{rms}} = -56.83 \text{ V}$$

$$(V_9)_{\text{rms}} = -14.65 \text{ V}$$

$$(V_{11})_{\text{rms}} = -23.156 \text{ V}$$

$$\therefore \text{T.H.D} = \frac{\sqrt{(V_3)_r^2 + (V_5)_r^2 + (V_7)_r^2 + (V_9)_r^2 + (V_{11})_r^2}}{(V_1)_r}$$

$$\text{THD} = 0.177941 \Rightarrow \underline{\underline{17.79\%}}$$

Q (a) no. of voltage levels per phase = $[2 \times [\text{no. of cells}]] + 1$
 $= (2 \times 3) + 1$
 $= 7$

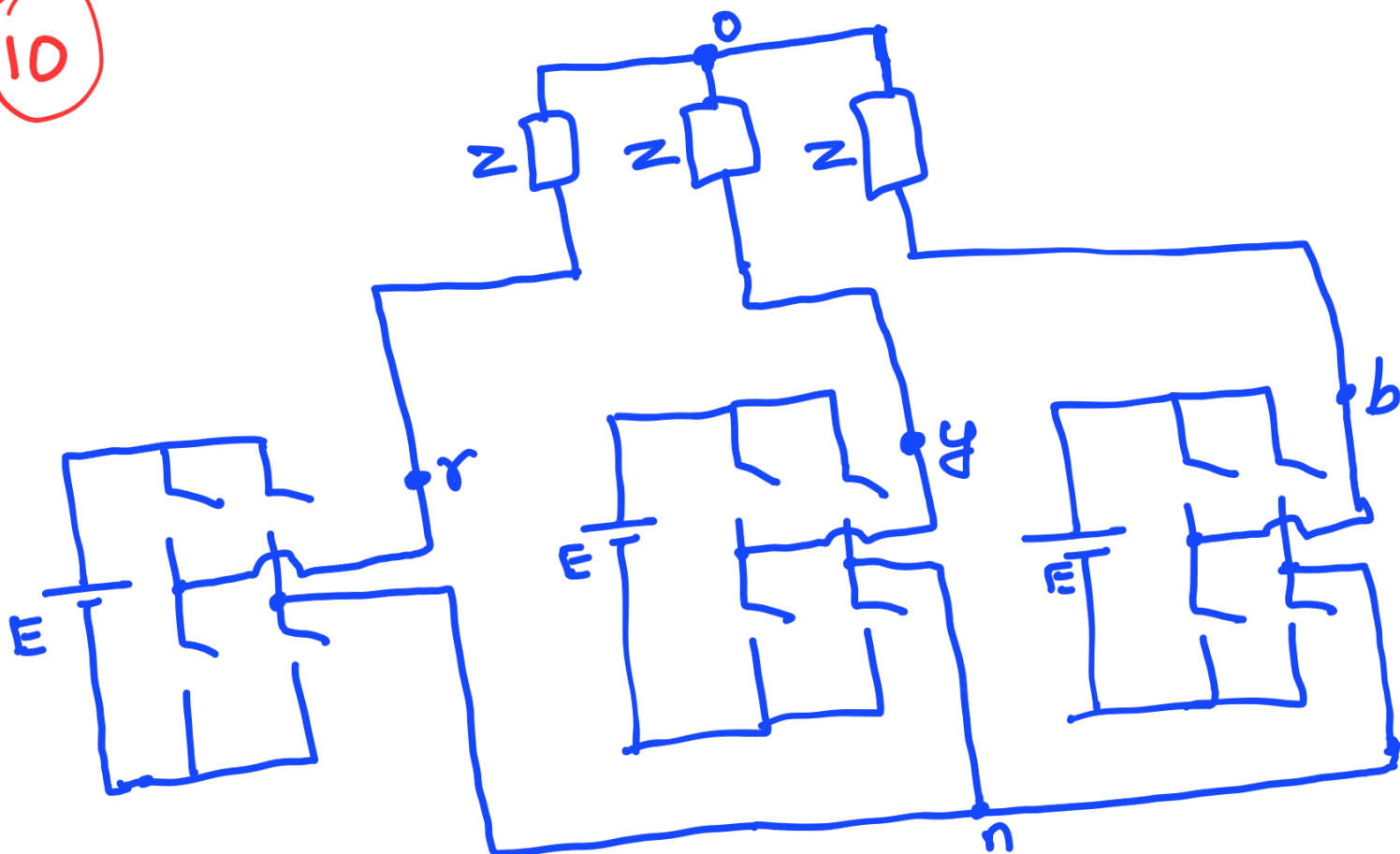
(b) single cell requires ^{minimum of} 4 IGBT and 4 diodes as, said in question that each phase is made up of 3 cells, therefore $3 \times [4+4] = 24$ devices per phase.

As, it is a 3-phase system, we would require a total of $3 \times 24 = \underline{\underline{72}}$ devices.

(c) $(m-1)$ carrier waves, when $m \rightarrow$ no. of levels.
 $(7-1) = 6$ carrier waves. $m = 2N+1$, $N \rightarrow$ no. of cells.

(d) $(2N+1)$ zero space vectors. $N =$ no. of cells
 $\therefore 7$ zero space vectors.

10



$$V_{rn} = V_{ro} + V_{on}$$

$$V_{yn} = V_{yo} + V_{on}$$

$$V_{bn} = V_{bo} + V_{on}$$

assuming

$$V_{ro} + V_{yo} + V_{bo} = 0 \rightarrow \text{balanced}$$

$$V_{on} = \frac{V_{rn} + V_{yn} + V_{bn}}{3}$$

$$\therefore V_{ro} = \frac{2}{3}V_{rn} - \frac{1}{3}V_{yn} - \frac{1}{3}V_{bn}$$

$$V_{yo} = -\frac{1}{3}V_{rn} + \frac{2}{3}V_{yn} - \frac{1}{3}V_{bn}$$

$$V_{bo} = -\frac{1}{3}V_{rn} - \frac{1}{3}V_{yn} + \frac{2}{3}V_{bn}$$

for switching state (0, -E, -E).

$$V_{rn} = 0, \quad V_{yn} = -E, \quad V_{bn} = -E$$

$$V_{ao} = \left[\frac{2}{3} \times 0 \right] - \left[\frac{1}{3} \times -E \right] - \left[\frac{1}{3} \times -E \right]$$

$$\boxed{V_{ao} = \frac{2E}{3}} \text{ "y," } \boxed{V_{go} = -\frac{E}{3}}, \text{ \& } \boxed{V_{bo} = -\frac{E}{3}}$$

$$V_{\alpha} = \sqrt{\frac{2}{3}} \left[V_{ao} - \frac{1}{2} V_{go} - \frac{1}{2} V_{bo} \right]$$

$$V_{\alpha} = \sqrt{\frac{2}{3}} \left[\frac{2E}{3} + \frac{E}{6} + \frac{E}{6} \right]$$

$$\boxed{V_{\alpha} = \sqrt{\frac{2}{3}} E}$$

$$V_{\beta} = \sqrt{\frac{2}{3}} \left[\frac{\sqrt{3}}{2} V_{go} - \frac{\sqrt{3}}{2} V_{bo} \right]$$

$$V_{\beta} = \sqrt{\frac{2}{3}} \cdot \frac{\sqrt{3}}{2} \left[-\frac{E}{3} - \left(-\frac{E}{3} \right) \right]$$

$$\boxed{V_{\beta} = 0}$$

$$\therefore \text{space vector, } V^* = V_{\alpha} + j V_{\beta}$$

$$V^* = \sqrt{\frac{2}{3}} E + j \cdot 0$$

$$\boxed{V^* = \sqrt{\frac{2}{3}} E \angle 0^\circ}$$

for switching state, $[E, 0, 0]$

$$V_{rn} = E, \quad V_{yn} = 0, \quad V_{bn} = 0$$

$$\boxed{V_{ro} = \frac{2}{3}E}, \quad \boxed{V_{yo} = -\frac{E}{3}}, \quad \boxed{V_{bo} = -\frac{E}{3}}$$

$$\therefore V_\alpha = \sqrt{\frac{2}{3}} \left[V_{ro} - \frac{V_{yo}}{2} - \frac{V_{bo}}{2} \right]$$

$$V_\alpha = \sqrt{\frac{2}{3}} \left[\frac{2}{3}E + \frac{E}{6} + \frac{E}{6} \right]$$

$$\boxed{V_\alpha = \sqrt{\frac{2}{3}}E}$$

$$V_\beta = \sqrt{\frac{2}{3}} \left[\frac{\sqrt{3}}{2} V_{yo} - \frac{\sqrt{3}}{2} V_{bo} \right]$$

$$V_\beta = \sqrt{\frac{2}{3}} \times \frac{\sqrt{3}}{2} \left[-\frac{E}{3} - \left(-\frac{E}{3}\right) \right]$$

$$\boxed{V_\beta = 0}$$

space vector, $\boxed{V^* = V_\alpha + j V_\beta = \sqrt{\frac{2}{3}}E \angle 0^\circ}$